

Air Quality Index (AQI)

What is the U.S. Air Quality Index (AQI)?

The U.S. AQI is EPA's index for reporting air quality.

How does the AQI work?

The U.S. Air Quality Index (AQI) is EPA's tool for communicating about outdoor air quality and health. The AQI includes six color-coded categories, each corresponding to a range of index values. The higher the AQI value, the greater the level of air pollution and the greater the health concern. For example, an AQI value of 50 or below represents good air quality, while an AQI value over 300 represents hazardous air quality.

For each pollutant an AQI value of 100 generally corresponds to an ambient air concentration that equals the level of the short-term national ambient air quality standard for protection of public health. AQI values at or below 100 are generally thought of as satisfactory. When AQI values are above 100, air quality is unhealthy: at first for certain sensitive groups of people, then for everyone as AQI values get higher.

The AQI is divided into six categories. Each category corresponds to a different level of health concern. Each category also has a specific color. The color makes it easy for people to quickly determine whether air quality is reaching unhealthy levels in their communities.

Daily AQI Color	Levels of concern	Values of Index	Description of Air Quality
Green	Good	0 to 50	Air quality is satisfactory, and air pollution poses little or no risk.
Yellow	Moderate	51 to 100	Air quality is acceptable. However, there may be a risk for some people, particularly those who are unusually sensitive to air pollution.
Orange	Unhealthy for Sensitive Groups	101 to 150	Members of sensitive groups may experience health effects. The general public is less likely to be affected.
Red	Unhealthy	151 to 200	Some members of the general public may experience health effects; members of sensitive groups may experience more serious health effects.
Purple	Very Unhealthy	201 to 300	Health alert: The risk of health effects is increased for everyone.
Maroon	Hazardous	301 and higher	Health warning of emergency conditions: everyone is more likely to be affected.

Research Questions

1. Which pollutants contribute most to Delhi's AQI?
2. How does AQI change across seasons?
3. Why is pollution higher during winter?
4. What geographical factors worsen Delhi's air quality?
5. What is the relationship between PM2.5 and PM10?
6. How does rainfall affect AQI?

Statistical Analysis of Delhi AQI

Descriptive Statistics : Descriptive statistics were used to summarize pollutant concentration levels in Delhi. Measures such as:

- Mean
- Minimum
- Maximum
- Standard deviation

2. AQI Breakpoint Analysis : AQI breakpoints were used to classify pollution levels into categories such as:

- Good
- Satisfactory
- Moderate
- Poor
- Very Poor
- Severe

The AQI values were calculated using CPCB breakpoint formulas.

PM2.5 and PM10 were observed as the major contributors to severe AQI conditions.

Example :

Let's take PM2.5 = 169.29

According to Indian AQI standards:

PM2.5 Range	AQI Range
121–250	301–400

So:

- $B_{LO} = 121$
 - $B_{HI} = 250$
 - $I_{LO} = 301$
 - $I_{HI} = 400$
-

Step 2: Apply Formula

AQI Formula:

$$I = \frac{(I_{HI} - I_{LO})}{(B_{HI} - B_{LO})} (C_p - B_{LO}) + I_{LO}$$

Substitute your values:

$$I = \frac{(400 - 301)}{(250 - 121)} (169.29 - 121) + 301$$

Step 3: Simplify

$$\begin{aligned} I &= \frac{99}{129} (48.29) + 301 \\ I &= 0.767 \times 48.29 + 301 \\ I &= 37.03 + 301 \\ I &\approx 338 \end{aligned}$$

Final Result

Pollutant	Value
PM2.5 Concentration	169.29

Pollutant	Value
AQI	338
Category	Very Poor

3. Correlation Analysis

Correlation analysis was performed to study relationships among pollutants.

Findings showed:

- Strong positive correlation between PM2.5 and PM10
- Similar pollution sources such as vehicular emissions, industrial activities, and construction dust

This helped identify interconnected pollution patterns in Delhi.

4. Seasonal Variation Analysis

The dataset was divided into:

- Winter
- Summer
- Monsoon
- Post-monsoon seasons

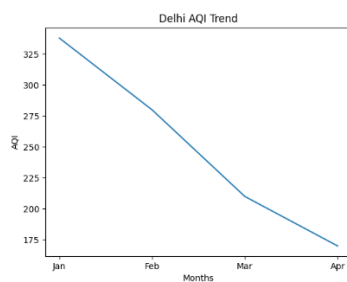
Analysis indicated:

- Highest AQI during winter and post-monsoon
- Lower AQI during monsoon due to rainfall and pollutant washout

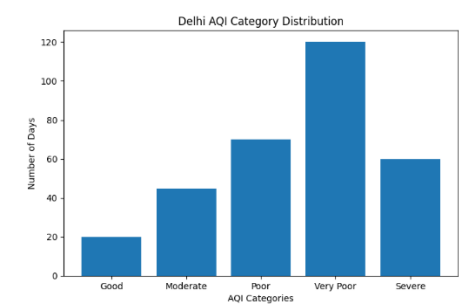
Seasonal changes significantly influence air quality levels in Delhi.

Visualizations

Line chart



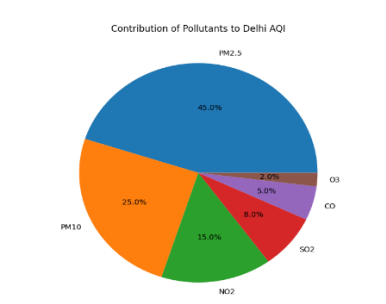
Bar chart:



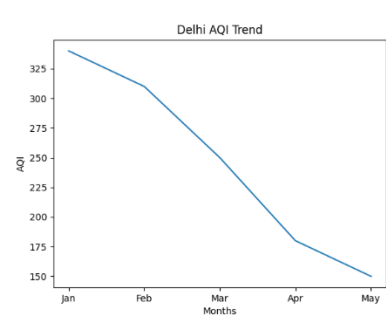
Heat map:

Pollutants	PM2.5	PM10	NO2	SO2	CO	O3
PM2.5	1.00	0.92	0.71	0.55	0.83	0.30
PM10	0.92	1.00	0.66	0.49	0.79	0.25
NO2	0.71	0.66	1.00	0.58	0.74	0.40
SO2	0.55	0.49	0.58	1.00	0.52	0.28
CO	0.83	0.79	0.74	0.74	1.00	0.35
O3	0.30	0.25	0.40	0.28	0.35	1.00

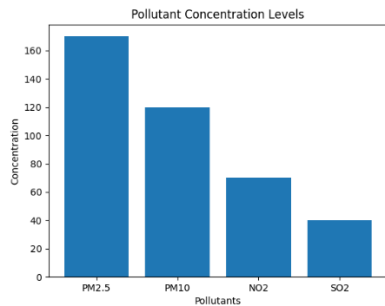
Pie chart:



AQI Trend Line Graph:



Pollutant Comparison Graph:



Environmental Challenges in Delhi

Major Challenges:

Vehicular Emissions:

Delhi has very high traffic density.

Construction Dust:

Large-scale urban construction increases PM10.

Crop Residue Burning:

Smoke from Punjab and Haryana affects Delhi heavily during October-November.

Industrial Pollution:

Factories around NCR emit harmful gases.

Geographical Factors:

Delhi is landlocked and pollution gets trapped because:

- Low wind speed
- Temperature inversion
- Dense urbanization

Weather Conditions:

Rainfall improves AQI temporarily by washing pollutants away.

Findings

- PM2.5 is the dominant pollutant in Delhi.
- Winter season records the highest AQI levels.
- Monsoon season shows cleaner air due to rainfall.

- AQI strongly correlates with particulate matter.
- Geographical conditions trap pollutants.

Recommendations:

- Promote electric vehicles
- Improve public transport
- Control construction dust
- Ban open waste burning
- Increase green cover
- Regional cooperation for stubble burning
- Public AQI awareness systems

Conclusion

- The analysis of Delhi's Air Quality Index (AQI) reveals that air pollution remains a major environmental and public health challenge in the city. Statistical analysis and visualizations showed that particulate matter pollutants, especially PM2.5 and PM10, are the primary contributors to poor air quality levels.
- Seasonal analysis indicated that AQI levels are highest during winter and post-monsoon periods due to temperature inversion, vehicular emissions, industrial activities, construction dust, and crop residue burning in neighboring regions. In contrast, monsoon seasons recorded comparatively lower AQI levels because rainfall helps reduce airborne pollutants.
- Correlation analysis demonstrated strong relationships between PM2.5, PM10, and other gaseous pollutants, suggesting common pollution sources such as traffic congestion and fossil fuel combustion. Geographical factors, including Delhi's landlocked location and limited wind circulation during winter, further intensify pollutant accumulation.
- The study highlights the urgent need for effective pollution control strategies such as promoting public transportation, reducing industrial emissions, controlling construction dust, increasing green cover, and improving regional coordination to manage stubble burning. Public awareness and continuous AQI monitoring are also essential for protecting human health and improving environmental sustainability in Delhi.